Explainable EfficientNetV2S Transfer Learning With an LLM-Assisted Reporting Module for Eight-Class Retinal OCT Disease Classification

1st Avinash Vadlamudi
MSc Data Analytics
Dublin City University
Dublin, Ireland
avinash.vadlamudi2@mail.dcu.ie

2nd Lohitha Vatti
MSc Data Analytics
Dublin City University
Dublin, Ireland
lohitha.vatti2@mail.dcu.ie

***Abstract*—Retinal disease screening using optical coherence tomography (OCT) can be delayed when image interpretation depends on specialist review. However, many existing deep-learning studies remain either too broad across unrelated imaging tasks or too narrow in disease coverage, and they often report accuracy without explaining which retinal regions influenced predictions. This study developed and evaluated a focused eight-class retinal OCT classification pipeline using the Retinal OCT-C8 dataset. The implementation used 24,000 OCT images from AMD, CNV, CSR, DME, DR, DRUSEN, MH, and NORMAL (seven pathological classes and one normal category), with balanced train, validation, and test partitions. Images were resized to 384×384 pixels and processed through an ImageNet-pretrained EfficientNetV2S model with controlled augmentation, transfer learning, fine-tuning of the final backbone layers, and explainable AI analysis using Grad-CAM, Grad-CAM++, and Eigen-CAM. The baseline model achieved 89.50% test accuracy, while fine-tuning increased test accuracy to 96.07% and macro F1-score to 0.961. Class-level analysis showed strong performance for AMD, CSR, DR, and MH, while CNV, DRUSEN and DME remained the main sources of confusion. The findings indicate that a compact, explainable EfficientNetV2S workflow can provide reproducible OCT-C8 disease classification while highlighting clinically relevant ambiguity rather than treating the model as a black box. Two reporting components were implemented, one structured per image (model output to structured clinical summaries via string formatting to a clinical knowledge base) and one free text (Groq/LLaMA-3.3-70B, temperature = 0.2). Clinical decisions are not taken by either component alone.**

***Index Terms*—Deep learning, EfficientNetV2S, explainable AI, Grad-CAM, Grad-CAM++, Eigen-CAM, optical coherence tomography, retinal disease classification, LLM-assisted reporting.**

I. Introduction

Optical coherence tomography (OCT) is a non-invasive retinal imaging technique which offers cross-sectional data on retinal layers and the structure of the macula. OCT plays a key role in the evaluation of retinal pathologies such as macular edema, choroidal neovascularization, drusen, age-related macular degeneration, macular hole and more, in retinal clinics. Automated classification of eight retinal OCT categories that might have similar texture, fluid, elevation or pattern of layer disruption is not general medical image classification and is the specific problem to which this study seeks to apply the framework [1], [2].

The difficulty is twofold. First, there have been few studies on public settings with more than four classes of interest, and the most common classes in public settings are CNV, DME, DRUSEN, and NORMAL. Second, medical image decision support requires the notion of classification accuracy, but not just that a model can fail due to the wrong reason, or be overconfident in a misclassifiable class pair. Explainable artificial intelligence (XAI) is thus crucial to reveal that the model focuses on reasonable retinal areas, instead of image backgrounds that are not relevant [3], [4].

The proposed system is not a substitute for ophthalmologists. Rather, it is intended as a research prototype that integrates classification, evaluation, error analysis and explanation. Clinical readiness cannot be demonstrated with an internal test-set; external, independent validation would be required along with expert ophthalmologist assessment, scanner type, hospital, acquisition setting and patient population calibration testing.

The challenge in this study is to develop an interpretable and focused deep learning pipeline to classify retinal OCT images into several disease categories. Automated classification can assist triage, but if it does not explain why a given classification was made, predictions may be seen as 'black-box' and diminish trust. Thus, in this study, we develop an EfficientNetV2S based model for retinal OCT classification into eight classes and utilise explainable AI to visualise the regions of the retina that affect the model's predictions.

The research question was: What degree of accuracy is achieved in identifying eight retinal categories from OCT images using an EfficientNetV2S-based explainable AI framework, and how can an LLM-Assisted reporting module provide transparent, clinically contextualised explanations from predictions?

The study has contributed: an eight-class OCT-C8 preprocessing and training protocol; a baseline and fine-tuned transfer learning comparison; class-wise error analysis for the main ambiguity of CNV, DME, DRUSEN and NORMAL;

XAI with Grad-CAM, Grad-CAM++ and Eigen-CAM; A per-image reporting function, which is deterministic, that gathers a structured summary through string-formatting model output with a clinical knowledge base, with no language model on this path, a reporting assistant (Groq/LLaMA-3.3-70B) to answer free-text questions about the assembled pipeline; a gap analysis between train and test accuracy at class level to measure class-level generalisation.

II. Literature Review and Gap Synthesis

A. OCT Disease Classification

The retinal OCT classification was a significant milestone when Kermany et al. achieved image classification with deep learning for treatable retinal diseases with CNV, DME, DRUSEN and NORMAL OCT images [2]. The idea was then expanded using residual networks [3], [4], [6], ensemble learning [9], hierarchical classification [5] and custom CNNs [8]. All of these studies found that transfer learning can achieve high accuracy when OCT images are organized into clinically significant disease categories. This body of literature is largely focused on four-class tasks, however, and many studies have not evaluated the reliability of the same pipeline when the class space has a broader range of retinal diseases. Recent OCT studies demonstrate that the quality of datasets, presence of label noise and the number of training images have significant impact on the reliability of classification [7]. Based on the literature, there is a need to use transfer learning and to report per-class rather than only global accuracy values.

B. Transfer Learning and Efficient Networks

EfficientNet introduced compound scaling, a technique that simultaneously scales network depth, width and input resolution [10], subsequently extended in EfficientNetV2 with Fused-MBConv blocks and training-aware architecture search [16]. EfficientNetV2S applies compound scaling simultaneously across depth, width, and resolution:

$$d = \alpha^{\varphi}, \quad w = \beta^{\varphi}, \quad r = \gamma^{\varphi} \tag{1}$$

$$\text{subject to: } \alpha \cdot \beta^2 \cdot \gamma^2 \approx 2,\ \alpha \geq 1,\ \beta \geq 1,\ \gamma \geq 1 \tag{2}$$

where φ is the compound coefficient and d, w, r are the depth, width and resolution scaling coefficients. EfficientNetV2S further substitutes depthwise separable convolutions with Fused-MBConv blocks at the beginning to boost GPU training efficiency. EfficientNetV2S was selected as it provides significantly stronger feature extraction (21.6M parameters vs 5.3M in EfficientNetB0) with Fused-MBConv blocks in early stages, while remaining feasible on a single GPU [16].

C. Explainability for OCT Decision Support

Grad-CAM produces a class-discriminative heatmap by weighting gradients flowing into the final convolutional layer [11]. Grad-CAM++ extends this using positive gradients and can work better in certain scenarios [12]. Eigen-CAM leverages principal components of activation maps to compute gradient-free explanations [13]. These techniques are useful for retinal OCT as they can verify that a model focuses on fluid pockets, drusen-like elevations, macular holes, or disruption of structure, instead of background regions.

The specific gap is the availability of a focused and reproducible workflow for retinal OCT comprising fine-tuned transfer learning, per-class performance analysis, confusion-pair interpretation, and multiple XAI approaches, across eight classes. The present paper fills that gap. He et al. suggested a model explainable using transformers on the same OCT-C8 dataset, proving that attention models can equal CNNs' efficiency [17]. Moreover, Guo et al. proved that softmax outputs are not calibrated probabilities and should be assessed by reliability diagrams and ECE [18].

III. Materials and Methods

The implementation was developed in Python using Google Colab/Renku, TensorFlow, Keras, NumPy, Pandas, Matplotlib, PIL, and scikit-learn. The overall flow of the proposed system is presented in Fig. 1.

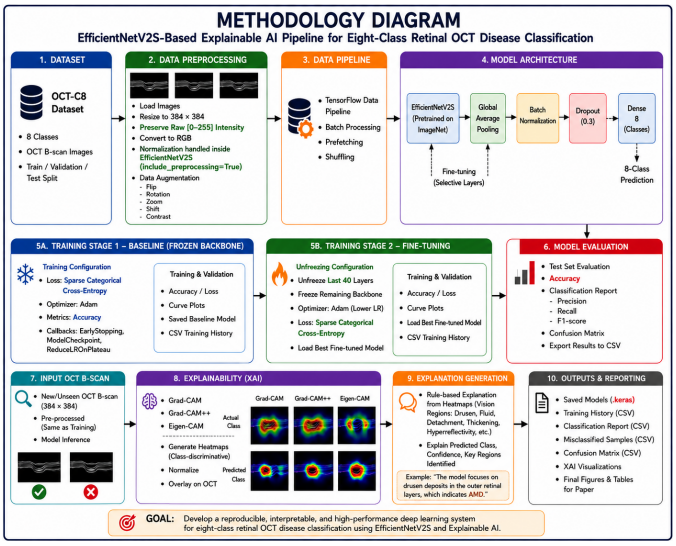


Fig. 1. Explainable AI pipeline for 8-class retinal OCT classification using EfficientNetV2S.

A. Dataset and Preprocessing

The study used the Retinal OCT-C8 dataset [1], a public dataset comprising 24,000 2D OCT images divided into 8 classes: AMD, CNV, CSR, DME, DR, DRUSEN, MH, and NORMAL. The dataset can be found with DOI 10.34740/KAGGLE/DSV/9595300, is licensed with CC BY-NC-SA 4.0. Documentation of the dataset publicly available was not detailed enough to provide information on the number of unique patients, original clinical institutions, scanner manufacturers, acquisition protocols, demographic composition, or annotation procedure. These data were not inferred, but recorded as unavailable. Patient-level separation between the training, validation and test partitions was not independently verified as patient identifiers were not available. The test folder is described as a held-out internal test set, not an independent

external test set. Each image is represented in the data set with one class label, which is an oversimplification of clinical reality because retinal abnormalities may occur in the same patient or scan. The training, validation and test images were 2,300, 350 and 350 per class respectively, as summarised in Table I. A balanced class design was used to limit re-sampling and facilitated interpretation at the macro level. All 24,000 images were subjected to an SHA-256 hash audit to find duplicate copies of them that were stored on different partitions. As summarised in Table II, 28 duplicate pairs were found between the training partition and validation partition and 19 duplicate pairs were found between the training partition and test partition. DRUSEN, CNV and DME were the only three classes in both directions that were contaminated. There were no duplicates found in the AMD, CSR, DR and MH or NORMAL groups. In total, 19 of 2,800 test images were contaminated by train–test contamination (0.68%). A further duplicate pair was identified within DRUSEN and the validation and test partitions. As neither partition is used for training, this does not constitute train–test leakage. Two of the train–validation pairs were images which were byte-identical but belonged to different classes (DRUSEN vs CNV), which suggested that there were some inconsistencies in the annotations at the DRUSEN - CNV boundary. Near-duplicates cannot be excluded as only exact SHA-256 matching was used in the audit.

TABLE I
OCT-C8 DATASET DISTRIBUTION

Class	Train	Val.	Test	Total
AMD	2,300	350	350	3,000
CNV	2,300	350	350	3,000
CSR	2,300	350	350	3,000
DME	2,300	350	350	3,000
DR	2,300	350	350	3,000
DRUSEN	2,300	350	350	3,000
MH	2,300	350	350	3,000
NORMAL	2,300	350	350	3,000
Total	18,400	2,800	2,800	24,000

Fig. 2 shows the image dimension distribution of the dataset before resizing. Width values varied from approximately 400 to 1,500 pixels, while height values were more concentrated between 300 and 600 pixels, justifying the standardisation to 384×384 pixels prior to training.

TABLE II
DUPLICATE IMAGES IDENTIFIED BY SHA-256 AUDIT.

Class	Train–Val	Train–Test	Val–Test	Test affected (%)
AMD	0	0	0	0.0
CNV	6	6	0	1.7
CSR	0	0	0	0.0
DME	8	2	0	0.6
DR	0	0	0	0.0
DRUSEN	12	11	1	3.1
MH	0	0	0	0.0
NORMAL	0	0	0	0.0
DRUSEN–CNV	2	0	0	0.0
Total	**28**	**19**	**1**	**0.68**

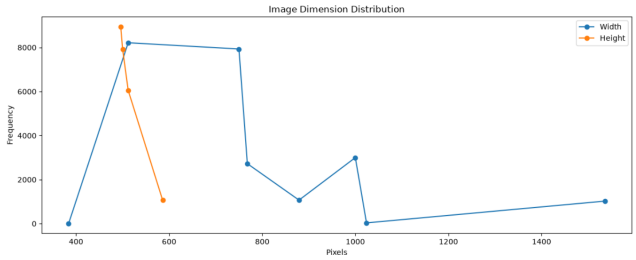


Fig. 2. Image dimension distribution of the OCT-C8 dataset showing width variability across samples. All images were resized to 384×384 pixels prior to training.

Images were resized to 384×384 pixels and loaded in chunks of 32 images. Implementations employed were based on TensorFlow/Keras in Google Colab with a fixed random seed of 42, deterministic operations where supported, and GPUs in mixed precision where available. Only clinically conservative transforms (horizontal flipping, small rotation, translation, zoom and contrast adjustment) were allowed for augmentation. The OCT images were loaded as grayscale images in RGB mode, with the grayscale images repeated in all three channels. Images were entered into the network as float32 tensors in the range $[0, 255]$ and all input scaling was done within the EfficientNetV2S built-in preprocessing layers $(include_preprocessing = True)$, which maps the input to $[-1, 1]$.



Fig. 3. Representative OCT images from the eight OCT-C8 classes used in the implementation. The figure signposts the focused retinal scope of the paper.

B. EfficientNetV2S Training Workflow

Table III shows the implementation protocol. The backbone is an ImageNet-pretrained EfficientNetV2S with the classification head removed. The model used global average pooling, batch normalisation, dropout (rate of 0.30), and an eight-unit softmax output layer.

The classification head produces class probabilities via softmax:

$$P(y = k \mid \mathbf{x}) = \frac{\exp(z_k)}{\sum_{j=1}^{8} \exp(z_j)}, \quad k \in \{1, \ldots, 8\} \qquad (3)$$

where z_k is the k-th logit from the Dense(8) output layer and the predicted class is $\hat{y} = \arg\max_k P(y = k \mid \mathbf{x})$.

Training minimised sparse categorical cross-entropy:

$$\mathcal{L} = -\sum_i \log P(y = y_i \mid \mathbf{x}_i) \tag{4}$$

where y_i is the integer ground-truth class label for sample i.

The backbone was frozen during baseline training and only the new classification head was trained. For fine-tuning, the final 40 backbone layers were made trainable while all BatchNormalisation layers remained frozen to prevent unstable statistics. The accuracy of validation, testing, class-wise precision, recall and F1-score were used to evaluate the model. Confusion matrices were used to identify dominant errors in the pair of diseases. Grad-CAM [11], Grad-CAM++ [12] and Eigen-CAM [13] maps were generated from correctly classified and misclassified samples. All three methods targeted the same layer, found programmatically using `find_last_4d_layer()`, which walks the EfficientNetV2S backbone backwards and returns the final layer producing a four-dimensional output tensor. This is equivalent to the `top_activation` layer, which has output dimensions of (batch, 12, 12, 1280) for 384×384 input images. The three methods were tested on a fixed stratified sample of 400 test images (50 per class) with measured run-time after a warm-up pass. A map was considered failed if it included NaN or a dynamic range less than 10^{-6}. The pre-softmax score was used to calculate gradients (with respect to). float32.

TABLE III
IMPLEMENTATION AND TRAINING PROTOCOL

Component	Configuration
Input	$384 \times 384 \times 3$ OCT image
Backbone	EfficientNetV2S, ImageNet weights, `include_top=False`
Head	GlobalAveragePooling2D, BatchNorm, Dropout(0.30), Dense(8, softmax, float32)
Augmentation	RandomFlip(horizontal), RandomRotation(0.03), RandomTranslation(0.03, 0.03), RandomZoom($\pm$0.05), RandomContrast(0.1)
Baseline	Adam, lr=1e-4, SparseCategoricalCrossentropy, 20 epochs
Fine-tuning	Last 40 backbone layers trainable, Adam lr=1e-5
Callbacks	ModelCheckpoint, ReduceLROnPlateau, EarlyStopping (patience=5), CSVLogger
Metrics	Accuracy, precision, recall, F1-score, confusion matrix
Environment	Google Colab/Renku, TensorFlow 2.20, mixed precision (`mixed_float16`); BatchNorm layers explicitly set `trainable=False` after `backbone.trainable=True` during fine-tuning. Baseline training ran for 20 epochs. Fine-tuning used EarlyStopping (patience=5); best checkpoint saved on validation accuracy improvement.

Grad-CAM++ computes the class activation map as:

$$L^c_{\text{Grad-CAM++}} = \text{ReLU}\left(\sum_k \alpha_k^c \cdot A^k\right) \tag{5}$$

where A^k is the k-th feature map of the `top_activation` layer and α_k^c are importance weights derived from second-order partial derivatives of the class score S^c.

The LLM-Assisted reporting component was implemented after the final-artifact export stage. It loads the saved `baseline_vs_finetuned_comparison.csv`, `finetuned_test_classification_metrics.csv` and `top_confusion_pairs.csv` files. Two separate functions were implemented: `predict_and_explain()` and `ask_reporting_agent()`. The `predict_and_explain()` function accepts a single OCT image as input, loads the fine-tuned EfficientNetV2S model, performs disease classification, generates Class Activation Mapping (CAM) heatmaps, retrieves the corresponding `CLINICAL_NOTES` entry, and constructs a clinician-friendly explanation using predefined string formatting rather than invoking a large language model (LLM). In contrast, the `ask_reporting_agent()` function processes the exported CSV files containing the prediction results and pipeline summaries to answer user queries regarding the completed analysis. Only structured textual summaries are forwarded to the Groq API, while the original OCT images and heatmaps are never transmitted. The reporting agent uses the `llama-3.3-70b-versatile` model with a temperature setting of `0.2` to produce concise and deterministic responses. A Groq API key is required for this component. The CLINICAL_NOTES were extracted from various ophthalmology literature sources and were not evaluated by an ophthalmologist. The system merely rephrases the predefined terms without developing its own diagnoses and treatments.

IV. RESULTS

A. Overall Model Performance

The learning curves in Fig. 4 show that the baseline model quickly converged, and the model was further fine-tuned to achieve better validation results. This improvement is confirmed by Table IV , where test accuracy increased from 89.50% (val: 89.36%, val loss: 0.2975) to 96.07% (val: 96.18%, val loss: 0.1225) after fine-tuning, while test loss decreased from 0.3047 to 0.1169. The 6.57 percentage-point increase is meaningful, as the split and evaluation method are identical for both stages. Wilson confidence intervals for the 2,800 predictions are [88.31%, 90.58%] for the baseline approach and [95.29%, 96.73%] for the fine-tuned network. These gaps are not overlapping, which is consistent with an actual improvement, but does not by itself establish statistical significance. A paired test such as McNemar's could not be performed because the per-image predictions from the baseline stage were not retained, so the comparison is unpaired.

The total number of parameters in EfficientNetV2S is 21.6M of which about 2.1M were trainable during fine-tuning (last 40 backbone layers).

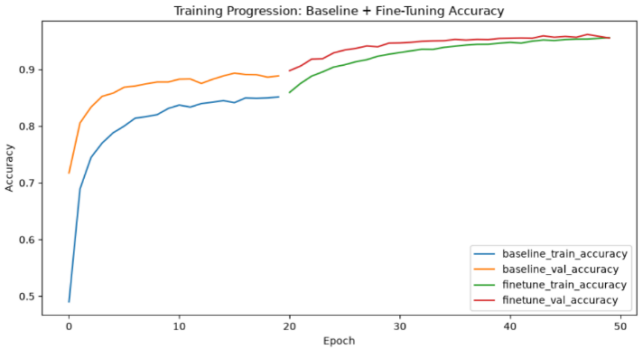


Fig. 4. Baseline and fine-tuned EfficientNetV2S training and validation accuracy curves.

TABLE IV
PERFORMANCE COMPARISON BETWEEN THE BASELINE AND FINE-TUNED EFFICIENTNETV2S MODELS.

Metric	Baseline	Fine-tuned
Validation Loss	0.2975	0.1225
Validation Accuracy	89.36%	96.18%
Test Loss	0.3047	0.1169
Test Accuracy	89.50%	96.07%
95% CI	88.31–90.58	95.29–96.73

B. Class-Wise Performance and Error Structure

The fine-tuned model achieved perfect F1-scores for AMD, CSR, DR, and MH (1.000). The weakest class was DRUSEN (F1 = 0.888, precision = 0.869), followed by CNV (F1 = 0.916) and DME (F1 = 0.934). The pattern is clinically similar to DRUSEN, with the potential for AI to diagnose DRUSEN based on OCT B-scan and human specialists to diagnose CNV based on subretinal fluid signatures, which is an area of overlap.

The macro-averaged F1-score across $C = 8$ classes is computed as:

$$F1_{\text{macro}} = \frac{1}{C}\sum_{c=1}^{C}\frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \tag{6}$$

where macro averaging weights all classes equally regardless of support, which is appropriate given the balanced class distribution of 350 test images per class.

TABLE V
FINE-TUNED EFFICIENTNETV2S CLASS-WISE TEST PERFORMANCE

Class	Precision	Recall	F1-Score
AMD	1.000	1.000	1.000
CNV	0.924	0.909	0.916
CSR	1.000	1.000	1.000
DME	0.955	0.914	0.934
DR	1.000	1.000	1.000
DRUSEN	0.869	0.909	0.888
MH	1.000	1.000	1.000
NORMAL	0.941	0.954	0.948
Macro avg.	**0.961**	**0.961**	**0.961**

Fig. 5 illustrates the per-class metrics visually. The bar chart confirms that AMD, CSR, DR and MH achieved perfect scores across all three metrics, while DRUSEN shows the lowest precision bar among all classes.

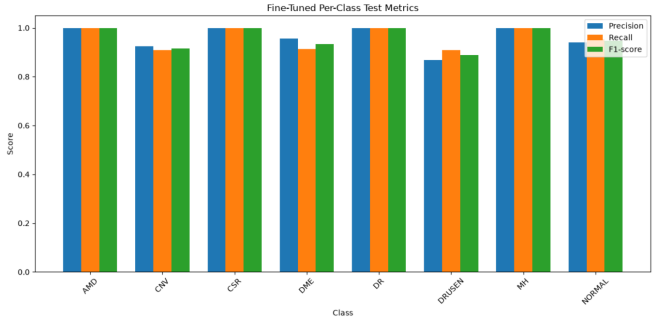


Fig. 5. Fine-tuned EfficientNetV2S per-class precision, recall and F1-score on the held-out test set. AMD, CSR, DR and MH achieved perfect scores; DRUSEN showed the lowest precision (0.869).

To quantify class-level generalisation, the per-class train-test accuracy gap was computed as:

$$\text{Gap}_c = \text{Acc}_{\text{train}}(c) - \text{Acc}_{\text{test}}(c) \tag{7}$$

The gap for all eight classes is summarised in Table VI. There was 0% gap and 100% test accuracy for four classes (AMD, CSR, DR and MH). This could indicate strong feature learning, but a 0% difference does not necessarily indicate that the model learned true retinal morphology, as dataset characteristics could also play a role. DME (8.1%), CNV (7.9%) and DRUSEN (7.5%) showed the largest gaps. This may be due to their shared subretinal fluid patterns; however, this is a hypothesis only and must await expert annotation or lesion-masked evaluation to be confirmed [17]. Gap values alone are not sufficient to determine the source of errors, whether due to clinical ambiguity or model limitation.

TABLE VI
PER-CLASS TRAIN-TEST ACCURACY GAP FOR FINE-TUNED EFFICIENTNETV2S

Class	Train Acc.	Test Acc.	Gap
AMD	100.0%	100.0%	0.0%
CNV	98.8%	90.9%	7.9%
CSR	100.0%	100.0%	0.0%
DME	99.5%	91.4%	8.1%
DR	100.0%	100.0%	0.0%
DRUSEN	98.4%	90.9%	7.5%
MH	100.0%	100.0%	0.0%
NORMAL	99.1%	95.4%	3.7%

Further description of these mistakes is presented in the confusion matrix in Fig. 6. Most AMD, CSR, DR, and MH images were correctly classified. The major confusion occurred between the following pairs: NORMAL-DRUSEN, DRUSEN-NORMAL, DME-DRUSEN, DRUSEN-DME, DME-NORMAL and CNV-DME.

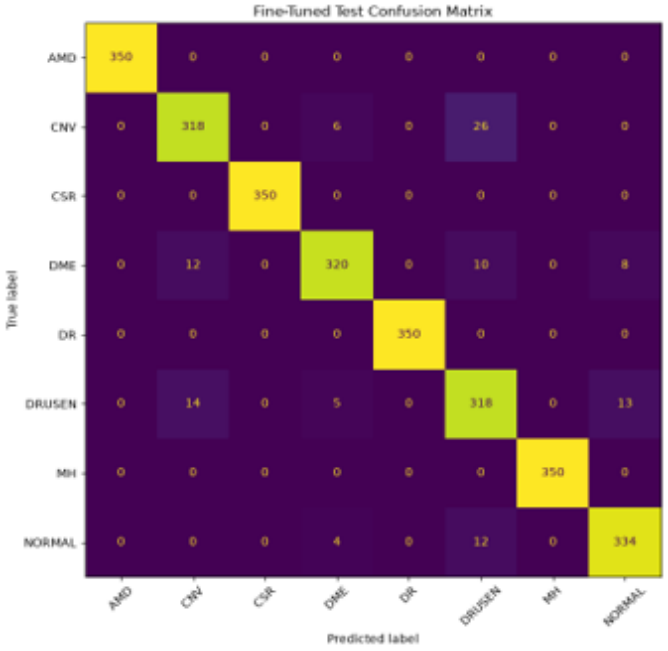


Fig. 6. Fine-tuned model confusion matrix on the held-out test set.

C. Explainable AI Analysis

A representative set of XAI panels is reported in Fig. 7. Examples of AMD, CNV and DME were correctly classified and highlighted active areas of the retina instead of the background of the image, but without validation against expert annotations or lesion masks [11]–[13]. A misclassified DRUSEN-to-NORMAL example illustrates why this matters: although the prediction was wrong, the heatmap continued to activate on genuine retinal tissue rather than the image background.

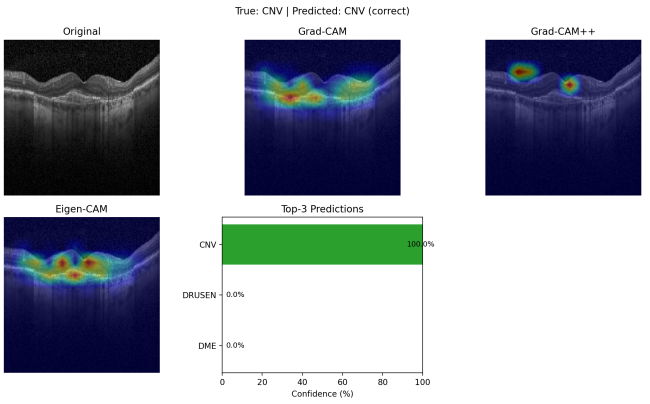


Fig. 7. Explainable AI example: True class: CNV, Predicted: CNV (correct, confidence: 100.0%) using Grad-CAM [11], Grad-CAM++ [12] and Eigen-CAM [13]. Original OCT image shown with three heatmap overlays. Highlighted regions indicate model attention and have not been validated against expert lesion annotations.

Fig. 8 shows a misclassification example where the model predicted DRUSEN with high confidence (1.000) for a CNV image. All three XAI methods highlight the subretinal region, yet the model still misclassified the image. This illustrates that high softmax confidence does not guarantee correct prediction, consistent with Guo et al. [18].

TABLE VII
EXPERIMENTAL COMPARISON OF XAI METHODS ON 400 STRATIFIED TEST IMAGES. A WARM-UP PASS IS EXCLUDED FROM RUNTIME. A MAP WAS CONSIDERED TO FAIL IF IT HAD A DYNAMIC RANGE BELOW 10^{-6} OR CONTAINED A NAN VALUE.

Method	Mean runtime (ms ± SD)	Failed maps (%)
Grad-CAM	1363.1 ± 55.9	0.0
Grad-CAM++	2827.3 ± 102.2	0.0
Eigen-CAM	827.0 ± 231.5	0.0

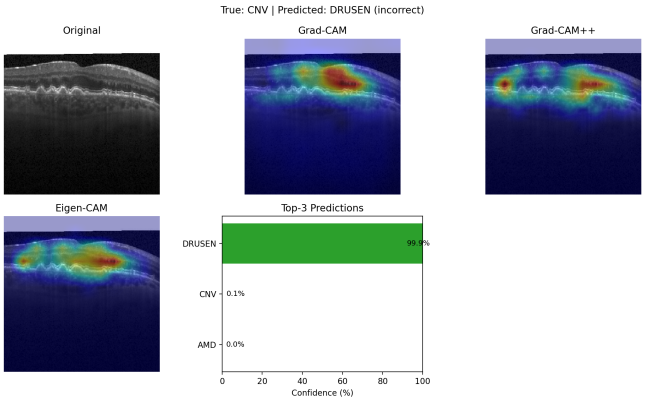


Fig. 8. XAI analysis for a misclassified example: True class: CNV, Predicted: DRUSEN (incorrect, confidence: 1.000). Grad-CAM, Grad-CAM++ and Eigen-CAM highlight the subretinal region yet the prediction was wrong. High softmax confidence does not guarantee correct prediction [18]. Highlighted regions have not been validated against expert annotations.

Table VII compares the three methods. All generated valid maps for each of the images. Eigen-CAM was the fastest method at 827.0 ms per image, since no backward pass is required, compared to 1363.1 ms for Grad-CAM and 2827.3 ms for Grad-CAM++, due to the nesting of the gradient tapes that results from its higher-order derivatives. In the initial implementation, the gradients computed with respect to the softmax output, as opposed to the pre-softmax score, resulted in invalid (NaN) maps for 11.5% of Grad-CAM and 20.8% of Grad-CAM++ outputs, only at high-confidence predictions (≥ 0.999), since the softmax gradient is zero when the score is saturated. These failures were resolved by using the pre-softmax score computed in float32, and all three methods yielded valid maps in this corrected version, whose results are presented in Table VII.

In Fig. 7, the original OCT image is displayed in each panel with three heatmap overlays. True class, predicted class, confidence and prediction outcome (correct/incorrect) are given for each example. The highlighted areas are those of the model's attention and are not verified with expert annotations of the lesions. The highlighted morphology was not distinct enough for the model to confidently separate subtle drusen-like change from a normal appearance, which is consistent

with the confusion pairs identified above.

D. LLM-Assisted Reporting Module Results

The LLM-Assisted reporting module was used after the final training phase. It utilises the `predict_and_explain()` function, which loads the saved `efficientnetv2s_finetuned_best.keras` model, computes Grad-CAM, Grad-CAM++ and Eigen-CAM heatmaps, retrieves structured clinical knowledge from the CLINICAL_NOTES knowledge base (8 entries), and assembles the summary by string formatting into a structured report containing diagnosis, confidence, area of the eye impacted, pattern and clinical notes. The generated reports are research demonstrations and have not been clinically validated by ophthalmologists or patients.

Fig. 9 shows an example output for an AMD prediction, in which the module retrieved the structured AMD CLINICAL_NOTES entry and populated each field of the report.

```
============================================================
  CLINICAL SUMMARY
============================================================
  Diagnosis    : AMD
  Confidence   : 100.0%
  Affected area: central retinal region
  Pattern      : minimal (0.0% of image activated)
  Clinical note: Age-related macular degeneration. Affects outer retina/RPE. Look for drusen, RPE irregularity, subretinal fluid.
============================================================
Agent >  Tell me about previous prediction

Agent answer:
Based on the project summary, I can provide an overview of the previous prediction results.

The model was trained on the Retinal OCT-C8 dataset, which consists of 24,000 images across 8 classes. After fine-tuning, the model achieved a high ac
curacy of 96.07% on the test set and 96.18% on the validation set.

In terms of class-wise performance, the model excelled in predicting certain classes, such as:

* AMD (Age-related Macular Degeneration)
* CSR (Central Serous Retinopathy)
* MH (Macular Hole)

These classes achieved a perfect F1 score of 1.000, indicating that the model was able to correctly classify all images in these classes.

On the other hand, the model struggled with the following classes:

* DRUSEN (Drusen): 0.888 F1 score
* CNV (Choroidal Neovascularization): 0.916 F1 score
* DME (Diabetic Macular Edema): 0.934 F1 score

The confusion matrix revealed that the model often confused CNV with DRUSEN, and vice versa. Additionally, the model sometimes misclassified DRUSEN as
NORMAL, and NORMAL as DRUSEN. There were also instances where DME was misclassified as CNV.
```

Fig. 9. Example LLM-Assisted reporting module output for an AMD prediction (confidence: 100.0%). The module retrieved the AMD CLINICAL_NOTES entry and the summary was formed using string formatting of the model output and not with the help of a language model. This is a research demonstration and has not been clinically validated.

Natural-language questions were used for the completed pipeline and were processed by the `ask_reporting_agent()` function. Questions that were matched in a predefined set of keywords were answered deterministically from the exported CSV files, and those that were not matched, were sent to the Groq API with a context block with the accuracy summary, class metrics and confusion pairs. Per-request latency was not consistently measured and there is no fallback; if the Groq API is not available an exception will be raised. For future work, there are latency profiling and graceful degradation.

When a DRUSEN image was misclassified as NORMAL (confidence 71.3%), Grad-CAM++ highlighted diffuse RPE activation rather than localised deposits, suggesting a possible failure mode rather than confirming it. The top-3 output (NORMAL 71.3%, DRUSEN 21.4%, AMD 5.1%) flagged the ambiguity. Although the softmax score was 71.3%, this is not necessarily an indication of lower confidence, as softmax scores are not calibrated probability estimates without further calibration analysis [18]. The top-3 outputted this ambiguity instead of presenting a single overconfident prediction.

The consistency was also checked for reporting outputs. The per-image module (`predict_and_explain()`) is deterministic – the diagnosis is the top-1 predicted class, confidence is the softmax output, and clinical note is the exact lookup from the dictionary. Thus, consistency follows from the construction.

The LLM path was evaluated on a subsample of 15 out of 240 test images, including both correct and misclassified predictions, as well as low-confidence predictions (Table VIII). For each output, we checked whether the predicted diagnosis and confidence coincide with the output of the model, whether all claims about the clinical status are linked to the corresponding `CLINICAL_NOTES` field, and whether the uncertainty is accounted for. Both diagnosis and confidence coincided in 15/15 cases; all 12/15 met all criteria. Three failed due to adding a finding that is not stated in the note, adding inappropriate reassurance (”Don’t worry”), and un-doing the hedge in the last sentence.

Another limitation that the review revealed and that cannot be described through the consistency scores is the fact that a DME image misclassified as NORMAL with 98.5% confidence resulted in the generation of the report that claimed that the retina was healthy and did not show any signs of pathology, whereas there was some disease. Similarly, another DME image misclassified as NORMAL with 62.7% confidence had DME as one of the possible alternatives in the predictions. Thus, one can conclude that the safety of reports depends on the confidence of the classifier, which is not the calibrated probability [18]. In other words, the reporting layer can strengthen the wrong predictions of the classifier. Such an assessment was made by the authors according to their observations.

V. Discussion

This study addresses several limitations common in the broader literature. The problem statement has been narrowed to eight-class retinal OCT classification. Literature synthesis is categorised by technical theme: OCT classification, efficient transfer learning, explainability, and remaining gap.

The reporting layer was used to enhance the practical value. The per-image summaries are generated deterministically by string formatting. The Groq/LLaMA-3.3-70B model is only used by the separate reporting assistant and only for questions not in its predetermined set of keywords. The module successfully converted saved metrics and CSV outputs into concise answers for project summary, dataset description, model performance, class difficulty and limitations. The LLM-Assisted reporting module is just a reporting layer - it does not alter predictions, retrain the model, or replace expert interpretation.

EfficientNetV2S can be trained with 24,000 images on a standard Renku/Colab GPU at 384×384 pixels, remaining much less demanding than larger CNNs or transformers. With 96.07% test accuracy, the result surpasses the EfficientNetV2S baseline (Stage 1, head-only training, 89.50%) by 6.57 percentage points. The 19 train–test duplicate images were re-

TABLE VIII
CONSISTENCY REVIEW OF GENERATED REPORTS (SAMPLE OF 15)

Case	True	Pred.	Conf.	Consistent with prediction
Correct, high conf.	DME	DME	1.000	No
Correct, high conf.	MH	MH	1.000	Yes
Correct, high conf.	MH	MH	1.000	Yes
Correct, high conf.	DR	DR	1.000	Yes
Correct, high conf.	DME	DME	0.956	Yes
Misclassified	DME	DRUSEN	0.431	Yes
Misclassified	DME	CNV	0.687	Yes
Misclassified	DME	NORMAL	0.985	Yes
Misclassified	DME	CNV	0.992	Yes
Misclassified	DME	NORMAL	0.627	Yes
Low confidence	DME	DME	0.644	Yes
Low confidence	CSR	CSR	0.597	Yes
Low confidence	DRUSEN	DRUSEN	0.668	No
Low confidence	DRUSEN	DME	0.530	Yes
Low confidence	DRUSEN	NORMAL	0.774	No
Consistent: 12/15				

moved and a new test set ($n = 2{,}781$) was used to re-evaluate the model. All 19 are classified correctly and so removing them from both the correct count (2,690 of 2,800) and the total gives deduplicated accuracy of $(2690 - 19)/2781 = 96.04\%$ which is within 0.1 percentage points of the reported accuracy, meaning that such a result cannot be attributed to train - test leakage. The model was well validated on the held-out test set but needs further testing on external sets before clinical use. DRUSEN (F1 = 0.888) and CNV (F1 = 0.916) are the most difficult classes; future work should focus on the CNV-DME-DRUSEN boundary with segmentation-assisted or attention-guided approaches.

The improvement from Stage 1 (89.50%) to Stage 2 (96.07%) is attributable to fine-tuning the final 40 backbone layers, which allowed the higher-level ImageNet features to adapt to OCT-specific retinal textures, rather than to any architectural property of EfficientNetV2S, which is identical across both stages. The DRUSEN class is the most difficult (precision = 0.869) due to the variability in deposit size and morphological similarity to early AMD.

Softmax confidence scores are not directly probability scores. Guo et al. proved that modern deep neural networks are often overconfident and that softmax scores give an inflated representation of actual correctness likelihood [18]. Reliability diagrams and ECE were not calculated in this work and should be conducted in future before applying confidence thresholds to uncertainty characterisation.

Both classification accuracy and explanation quality are important - a plausible heatmap cannot compensate for an incorrect prediction [15]. Grad-CAM [11], Grad-CAM++ [12] and Eigen-CAM [13] provide complementary post-hoc views of model attention, but heatmap localisation quality must be assessed through expert annotation or lesion-mask evaluation before clinical validation.

Retinal OCT-C8 does not include patient metadata such as age, gender, ethnicity, scanning device model, or clinical centre. Demographic and device-specific performance differences cannot therefore be assessed. Future work should evaluate the pipeline on datasets with demographic and scanner metadata available [17].

VI. CONCLUSION

A focused, reproducible and explainable pipeline for retinal OCT disease classification was developed in this study. It addressed previous shortcomings by narrowing scope to eight-class OCT-C8 classification, restructuring the literature into technical themes, and using EfficientNetV2S transfer learning, fine-tuning, quantitative evaluation, misclassification analysis, and three XAI techniques.

The paper contains the dataset distribution table, representative OCT samples, training curves, baseline vs fine-tuned comparison table, the held-out test confusion matrix, the per-class metric table, selected XAI overlays, and the LLM-Assisted reporting module write-up. The LLM-Assisted reporting module is presented as a decision support tool that brings black-box predictions to light and places them in a clinical context, not as an independent diagnostic system.

The results support implementation of retinal screening support in education, telemedicine, and primary-care research. The model should not be used clinically without further validation. Patient-level separation was impossible to verify, demographic and scanner metadata were unavailable, and model calibration could not be tested - all prerequisites for clinical use. Future work should focus on independent OCT datasets, ophthalmologist validation, calibration assessment, and comparisons with Vision Transformers [14] and lightweight mobile CNNs.

GENERATIVE AI DISCLOSURE

Generative AI tools were used only to help improve language proficiency, check grammar, debug code, and clarify concepts.

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